

Multiple Linear Regression

Notes by Moutushi Chatterji L^AT_EXed by Sathvik H S

Introduction

1. Multiple Regression is appropriate when research problem involves a single dependent variable presumed to be related to two or more independent variables.
2. The objective is to predict changes in dependent variable in response to changes in independent variables.

Some Examples

- While estimating / predicting amount of a particular crop production in a year, one should take into account the effect of amount of rainfall, average humidity, natural soil fertility and so on.
- Monthly expenditure on dining out (DV) can be predicted from information regarding Family Income, Family Size and Age of the head of the family.
- Company's sales can be predicted from information on its expenditure on ads, number of sales people and number of retail outlets carrying its product.

Derivation

Suppose we have p variables $x_1, x_2, \dots, x_p$ and there are n values collected on each of the p variables as $(x_{1\alpha}, x_{2\alpha}, \dots, x_{p\alpha})$, for $\alpha = 1, 2, \dots, n$.¹

Here our objective is to predict any one of these p variables on the basis of the rest.

Without loss of generality, let us suppose that we want to estimate or predict x_1 in terms of $x_2, x_3, \dots, x_p$. So x_1 must be expressed as some mathematical function of $x_2, x_3, \dots, x_p$.

For our purpose, we only restrict ourselves to the linear mathematical function of $x_2, x_3, \dots, x_p$ like

$$x_1 = a_0 + a_2(x_2 - \bar{x}_2) + a_3(x_3 - \bar{x}_3) + \dots + a_p(x_p - \bar{x}_p) \quad (1)$$

where $\bar{x}_i = \frac{1}{n} \sum_{\alpha=1}^n x_{i\alpha}$ for $i = 2, 3, \dots, p$ and $a_0, a_2, a_3, \dots, a_p$ are unknown constants which need to be estimated on the basis of the given data with the help of *method of least squares*.

We assume the following linear model for the purpose of the estimation of the unknown constants $a_0, a_2, \dots, a_p$:

$$x_1 = X_{1.23\dots p} + e_{1.23\dots p} \quad (2)$$

¹Often the shorthand $1(1)n$ is used to represent the sequence $1, 2, \dots, n$ with the general syntax for shorthand being **start(step)end**. We will stick with the latter notation throughout.

$$\Rightarrow x_{1\alpha} = X_{1.23\dots p\alpha} + e_{1.23\dots p\alpha}, \alpha = 1, 2, \dots, n \quad (3)$$

Here,

$$X_{1.23\dots p\alpha} = a_0 + a_2(x_{2\alpha} - \bar{x}_2) + a_3(x_{3\alpha} - \bar{x}_3) + \dots + a_p(x_{p\alpha} - \bar{x}_p) \quad (4)$$

and $e_{1.23\dots p\alpha}$ is the error involved in estimating the α^{th} value of x_1 on the basis of $x_2, x_3, \dots, x_p$.

We now follow the method of least squares to estimate the unknown constants $a_0, a_2, a_3, \dots, a_p$.

As the name suggests, we want to minimize

$$\Delta_1 = \sum_{\alpha=1}^n e_{1.23\dots p\alpha}^2 = \sum_{\alpha=1}^n \{x_{1\alpha} - a_0 - a_2(x_{2\alpha} - \bar{x}_2) - a_3(x_{3\alpha} - \bar{x}_3) - \dots - a_p(x_{p\alpha} - \bar{x}_p)\}^2 \quad (5)$$

with respect to the constants $a_0, a_2, a_3, \dots, a_p$.

Derivation Using the Dispersion Matrix

Now, $\frac{\partial \Delta_1}{\partial a_0} = -2 \sum_{\alpha=1}^n \{x_{1\alpha} - a_0 - a_2(x_{2\alpha} - \bar{x}_2) - a_3(x_{3\alpha} - \bar{x}_3) - \dots - a_p(x_{p\alpha} - \bar{x}_p)\}$

Setting $\frac{\partial \Delta_1}{\partial a_0} = 0$, we get

$$\begin{aligned} 0 &= \sum_{\alpha=1}^n \{x_{1\alpha} - a_0 - a_2(x_{2\alpha} - \bar{x}_2) - a_3(x_{3\alpha} - \bar{x}_3) - \dots - a_p(x_{p\alpha} - \bar{x}_p)\} \\ \Rightarrow a_0 &= \frac{1}{n} \sum_{\alpha=1}^n \{x_{1\alpha} - a_2(x_{2\alpha} - \bar{x}_2) - a_3(x_{3\alpha} - \bar{x}_3) - \dots - a_p(x_{p\alpha} - \bar{x}_p)\} \\ \Rightarrow a_0 &= \frac{1}{n} \sum_{\alpha=1}^n x_{1\alpha} = \bar{x}_1^2 \end{aligned} \quad (6)$$

Now, for any $k = 2, 3, \dots, p$,

$$\frac{\partial \Delta_1}{\partial a_k} = -2 \sum_{\alpha=1}^n \{x_{1\alpha} - \bar{x}_1 - a_2(x_{2\alpha} - \bar{x}_2) - a_3(x_{3\alpha} - \bar{x}_3) - \dots - a_p(x_{p\alpha} - \bar{x}_p)\} (x_{k\alpha} - \bar{x}_k)$$

Note that, the value of a_0 is substituted as $\bar{x}_1$ in the above expression. Setting $\frac{\partial \Delta_1}{\partial a_k} = 0$ gives us

$$\begin{aligned} \sum_{\alpha=1}^n \{x_{1\alpha} - \bar{x}_1 - a_2(x_{2\alpha} - \bar{x}_2) - a_3(x_{3\alpha} - \bar{x}_3) - \dots - a_p(x_{p\alpha} - \bar{x}_p)\} (x_{k\alpha} - \bar{x}_k) &= 0 \\ \Rightarrow \sum_{\alpha=1}^n (x_{1\alpha} - \bar{x}_1) (x_{k\alpha} - \bar{x}_k) &= \sum_{j=2}^p a_j \sum_{\alpha=1}^n (x_{j\alpha} - \bar{x}_j) (x_{k\alpha} - \bar{x}_k) \\ \Rightarrow s_{1k} &= \sum_{j=2}^p a_j s_{jk}, \quad k = 2, 3, \dots, p \end{aligned} \quad (7)$$

where $s_{ij} = \sum_{\alpha=1}^n (x_{i\alpha} - \bar{x}_i) (x_{j\alpha} - \bar{x}_j)$ for every $i, j = 1, 2, \dots, p$.

²Since $\sum_{\alpha=1}^n (x_{i\alpha} - \bar{x}_i) = 0, i = 2, 3, \dots, p$. This is true in general, i.e., *sum of deviation about the mean* is always zero and we will use this fact frequently.

The collection of equations (7),³

$$\begin{aligned} s_{12} &= a_2 s_{22} + a_3 s_{32} + \dots + a_p s_{p2} \\ s_{13} &= a_2 s_{23} + a_3 s_{33} + \dots + a_p s_{p3} \\ &\vdots \quad \quad \quad \vdots \quad \quad \quad \ddots \\ s_{1p} &= a_2 s_{2p} + a_3 s_{3p} + \dots + a_p s_{pp} \end{aligned}$$

can be written concisely in matrix for as

$$\begin{pmatrix} s_{12} \\ s_{13} \\ \vdots \\ s_{1p} \end{pmatrix} = \begin{pmatrix} s_{22} & s_{32} & \dots & s_{p2} \\ s_{23} & s_{33} & \dots & s_{p3} \\ \vdots & \vdots & \ddots & \vdots \\ s_{2p} & s_{3p} & \dots & s_{pp} \end{pmatrix} \begin{pmatrix} a_2 \\ a_3 \\ \vdots \\ a_p \end{pmatrix} \Rightarrow \mathbf{s}_1 = S^* \mathbf{a} \quad (8)$$

With the assumption of non-singularity of the distribution, we have $|S^*| \neq 0$, since S^* is equivalent to the dispersion matrix of $x_2, x_3, \dots, x_p$.

Now, applying *Cramer's rule*, we get

$$\hat{a}_j = \frac{1}{|S^*|} \begin{vmatrix} s_{22} & \dots & s_{j-1,2} & s_{12} & s_{j+1,2} & \dots & s_{p2} \\ s_{23} & \dots & s_{j-1,3} & s_{13} & s_{j+1,3} & \dots & s_{p3} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ s_{2p} & \dots & s_{j-1,p} & s_{1p} & s_{j+1,p} & \dots & s_{pp} \end{vmatrix}$$

Now let us define the corrected SS-SP matrix⁴ of $x_1, x_2, x_3, \dots, x_p$ as

$$S = \begin{pmatrix} s_{11} & s_{21} & \dots & s_{p1} \\ s_{12} & s_{22} & \dots & s_{p2} \\ \vdots & \vdots & \ddots & \vdots \\ s_{1p} & s_{2p} & \dots & s_{pp} \end{pmatrix}$$

where $s_{ij} = \sum_{\alpha=1}^n (x_{i\alpha} - \bar{x}_i)(x_{j\alpha} - \bar{x}_j)$ for every $i, j = 1, 2, \dots, p$ (as defined previously). Also let S_{ij} be the cofactor of S corresponding to s_{ij} . Then,

$$S_{11} = (-1)^{1+1} \begin{vmatrix} s_{22} & s_{32} & \dots & s_{p2} \\ s_{23} & s_{33} & \dots & s_{p3} \\ \vdots & \vdots & \ddots & \vdots \\ s_{2p} & s_{3p} & \dots & s_{pp} \end{vmatrix} = |S^*|$$

Now,

$$\begin{vmatrix} s_{22} & \dots & s_{j-1,2} & s_{12} & s_{j+1,2} & \dots & s_{p2} \\ s_{23} & \dots & s_{j-1,3} & s_{13} & s_{j+1,3} & \dots & s_{p3} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ s_{2p} & \dots & s_{j-1,p} & s_{1p} & s_{j+1,p} & \dots & s_{pp} \end{vmatrix} = (-1)^{j-2} \begin{vmatrix} s_{12} & s_{22} & \dots & s_{j-1,2} & s_{j+1,2} & \dots & s_{p2} \\ s_{13} & s_{23} & \dots & s_{j-1,3} & s_{j+1,3} & \dots & s_{p3} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ s_{1p} & s_{2p} & \dots & s_{j-1,p} & s_{j+1,p} & \dots & s_{pp} \end{vmatrix}$$

³These equations are collectively called *the normal equations*. They will be used extensively while deriving extra results/observations.

⁴Sum of Squares-Sum of Products matrix

$$= (-1)^{j-2}(-1)^{1+j}S_{1j} = -S_{1j}$$

The first equality above follows as we need $j-2$ column swaps to move s_1 to the left. The second equality follows since the determinant is exactly equal to $(-1)^{1+j}S_{1j}$.⁵

Therefore, we have

$$\hat{a}_j = -\frac{S_{1j}}{S_{11}}, \quad j = 2, 3, \dots, p \quad (9)$$

So the multiple linear regression equation of x_1 on $x_2, x_3, \dots, x_p$ is,

$$\begin{aligned} x_{1.23\dots p} &= \bar{x}_1 - \frac{S_{12}}{S_{11}}(x_2 - \bar{x}_2) - \frac{S_{13}}{S_{11}}(x_3 - \bar{x}_3) - \dots - \frac{S_{1p}}{S_{11}}(x_p - \bar{x}_p) \\ \Rightarrow x_{1.23\dots p} &= \bar{x}_1 - \sum_{j=2}^p \frac{S_{1j}}{S_{11}}(x_j - \bar{x}_j) \end{aligned} \quad (10)$$

If we denote $x_{1.23\dots p}$ as x_1 , we get another form of the above equation:

$$\begin{aligned} -\sum_{j=2}^p \frac{S_{1j}}{S_{11}}(x_j - \bar{x}_j) &= x_1 - \bar{x}_1 \\ \Rightarrow \sum_{j=1}^p S_{1j}(x_j - \bar{x}_j) &= 0 \end{aligned} \quad (11)$$

Alternative Derivation Using the Correlation Matrix

Let r_{ij} be the simple product moment correlation coefficient between the variables x_i and x_j for every $i, j = 1, 2, \dots, p$. Note that $r_{ii} = 1$ for every $i = 1, 2, \dots, p$.

We also know that, $r_{ij} = \frac{s_{ij}}{s_i s_j}$ for every $i, j = 1, 2, \dots, p$ where $s_i^2 = s_{ii} = \sum_{\alpha=1}^n (x_{i\alpha} - \bar{x}_i)^2$.

$$\Rightarrow s_{ij} = r_{ij} s_i s_j; \quad i, j = 1, 2, \dots, p \quad (12)$$

Thus the cofactor of with respect to the element s_{ij} of the matrix S is

$$\begin{aligned} S_{ij} &= (-1)^{i+j} \begin{vmatrix} s_{11} & s_{21} & \cdots & s_{j-1,1} & s_{j+1,1} & \cdots & s_{p1} \\ s_{12} & s_{22} & \cdots & s_{j-1,2} & s_{j+1,2} & \cdots & s_{p2} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ s_{1,i-1} & s_{2,i-1} & \cdots & s_{j-1,i-1} & s_{j+1,i-1} & \cdots & s_{p,i-1} \\ s_{1,i+1} & s_{2,i+1} & \cdots & s_{j-1,i+1} & s_{j+1,i+1} & \cdots & s_{p,i+1} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ s_{1p} & s_{2p} & \cdots & s_{j-1,p} & s_{j+1,p} & \cdots & s_{pp} \end{vmatrix} \\ &= (-1)^{i+j} s_1^2 \cdots s_{i-1}^2 s_i s_{i+1}^2 \cdots s_{j-1}^2 s_j s_{j+1}^2 \cdots s_p^2 \begin{vmatrix} 1 & r_{21} & \cdots & r_{j-1,1} & r_{j+1,1} & \cdots & r_{p1} \\ r_{12} & 1 & \cdots & r_{j-1,2} & r_{j+1,2} & \cdots & r_{p2} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ r_{1,i-1} & r_{2,i-1} & \cdots & r_{j-1,i-1} & r_{j+1,i-1} & \cdots & r_{p,i-1} \\ r_{1,i+1} & r_{2,i+1} & \cdots & r_{j-1,i+1} & r_{j+1,i+1} & \cdots & r_{p,i+1} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ r_{1p} & r_{2p} & \cdots & r_{j-1,p} & r_{j+1,p} & \cdots & 1 \end{vmatrix} \end{aligned} \quad (13)$$

⁵ S_{1j} was defined to be equal to $(-1)^{1+j}$ times the determinant of interest.

If we take the correlation matrix to be R i.e,

$$R = \begin{pmatrix} 1 & r_{21} & \dots & r_{p1} \\ r_{12} & 1 & \dots & r_{p2} \\ \vdots & \vdots & \ddots & \vdots \\ r_{1p} & r_{2p} & \dots & 1 \end{pmatrix}$$

and take R_{ji} to be the cofactor of $(i, j)^{\text{th}}$ element of R , then we have

$$S_{ij} = (-1)^{i+j} s_1^2 \dots s_{i-1}^2 s_i s_{i+1}^2 \dots s_{j-1}^2 s_j s_{j+1}^2 \dots s_p^2 (-1)^{i+j} R_{ij}; i, j = 1, 2, \dots, p \quad (14)$$

This gives

$$\hat{a}_j = -\frac{S_{1j}}{S_{11}} = -\frac{s_1}{s_j} \cdot \frac{R_{1j}}{R_{11}}, j = 2, 3, \dots, p \quad (15)$$

Hence, the multiple linear regression equation now becomes

$$\begin{aligned} x_{1.23\dots p} &= \bar{x}_1 - \frac{s_1 R_{12}}{s_2 R_{11}} (x_2 - \bar{x}_2) - \frac{s_1 R_{13}}{s_3 R_{11}} (x_3 - \bar{x}_3) - \dots - \frac{s_1 R_{1p}}{s_p R_{11}} (x_p - \bar{x}_p) \\ &= \bar{x}_1 - \sum_{j=2}^p \frac{s_1 R_{1j}}{s_j R_{11}} (x_j - \bar{x}_j) \end{aligned} \quad (16)$$

Key Properties and Remarks

1. Usually, $\hat{a}_j$ is denoted as $b_{1j.2,\dots,j-1,j+1,\dots,p}$, where $b_{1j.2,\dots,j-1,j+1,\dots,p}$ is called *partial regression coefficient* of x_1 on x_j when other variables are kept fixed. The simple reason for this is

$$\frac{\partial X_{1.23\dots p}}{\partial x_j} = a_j$$

In other words, x_1 increases a_j units for unit increase in x_j when other variables are kept fixed.

2. Predicted mean of x_1 is equal to the observed mean of x_1 .

$$\bar{X}_{1.23\dots p} = \bar{x}_1 \quad (17)$$

Proof.

$$\begin{aligned} X_{1.23\dots p\alpha} &= \bar{x}_1 + \hat{a}_2 (x_{2\alpha} - \bar{x}_2) + \hat{a}_3 (x_{3\alpha} - \bar{x}_3) + \dots + \hat{a}_p (x_{p\alpha} - \bar{x}_p) \\ \Rightarrow \frac{1}{n} \sum_{\alpha=1}^n X_{1.23\dots p\alpha} &= \frac{1}{n} \sum_{\alpha=1}^n \bar{x}_1 + \hat{a}_2 \frac{1}{n} \sum_{\alpha=1}^n (x_{2\alpha} - \bar{x}_2) + \dots + \hat{a}_p \frac{1}{n} \sum_{\alpha=1}^n (x_{p\alpha} - \bar{x}_p) \\ &\Rightarrow \bar{X}_{1.23\dots p} = \bar{x}_1 \end{aligned}$$

□

3. The mean of residuals or equivalently sum of residuals is zero.

$$\bar{e}_{1.23\dots p} = 0 \quad (18)$$

Proof.

$$\begin{aligned}
x_1 &= X_{1.23\dots p} + e_{1.23\dots p} \\
\Rightarrow \bar{x}_1 &= \bar{X}_{1.23\dots p} + \bar{e}_{1.23\dots p} \\
\Rightarrow \bar{e}_{1.23\dots p} &= 0
\end{aligned}
\tag{from (17)}$$

□

Sum of Squares of Errors (SSE)

By definition we know that,

$$SSE = \min_{a_0, a_2, \dots, a_p} \Delta_1 = \sum_{\alpha=1}^n \{x_{1\alpha} - \hat{a}_0 - \hat{a}_2(x_{2\alpha} - \bar{x}_2) - \hat{a}_3(x_{3\alpha} - \bar{x}_3) - \dots - \hat{a}_p(x_{p\alpha} - \bar{x}_p)\}^2$$

Therefore, we can derive the expression for SSE as follows:

$$\begin{aligned}
SSE &= \sum_{\alpha=1}^n \left\{ x_{1\alpha} - \bar{x}_1 - \sum_{j=2}^p \hat{a}_j (x_{j\alpha} - \bar{x}_j) \right\} \left\{ x_{1\alpha} - \bar{x}_1 - \sum_{k=2}^p \hat{a}_k (x_{k\alpha} - \bar{x}_k) \right\} \\
&= \sum_{\alpha=1}^n \left[(x_{1\alpha} - \bar{x}_1)^2 - \sum_{k=2}^p \hat{a}_k (x_{1\alpha} - \bar{x}_1) (x_{k\alpha} - \bar{x}_k) - \sum_{j=2}^p \hat{a}_j (x_{j\alpha} - \bar{x}_j) \left\{ x_{1\alpha} - \bar{x}_1 - \sum_{k=2}^p \hat{a}_k (x_{k\alpha} - \bar{x}_k) \right\} \right] \\
&= s_1^2 - \sum_{k=2}^p \hat{a}_k s_{1k} - \sum_{j=2}^p s_p \hat{a}_j \left\{ s_{1j} - \sum_{k=2}^p \hat{a}_k s_{jk} \right\} \\
&= s_1^2 - \sum_{k=2}^p \hat{a}_k s_{1k} \tag{from (7)} \\
&= s_1^2 + \sum_{k=2}^p \frac{s_1 R_{1k}}{s_k R_{11}} \cdot r_{1k} s_1 s_k \tag{from (12) and (15)} \\
&= \frac{s_1^2}{R_{11}} \left(1 \cdot R_{11} + \sum_{k=2}^p r_{1k} R_{1k} \right) = \frac{s_1^2}{R_{11}} \left(r_{11} \cdot R_{11} + \sum_{k=2}^p r_{1k} R_{1k} \right) \tag{since } r_{11} = 1 \\
&= \frac{s_1^2}{R_{11}} \left(\sum_{k=1}^p r_{1k} R_{1k} \right)
\end{aligned}$$

Since the summation on right-hand side is simply determinant of the matrix R expanded along the first column, we get the final expression as

$$SSE = \frac{s_1^2}{R_{11}} |R| \tag{19}$$

Correlation Coefficient in Higher Dimension

- In biological, physical and social sciences, often data are available on more than two variables and value of one variable seems to be influenced by more than one variables.

- For example, crimes in a city may be influenced by illiteracy, increased population and unemployment in the city, etc.
- The production of a crop may depend upon amount of rainfall, quality of seeds, quantity of fertilizers used and method of irrigation, etc.
- Similarly, performance of students in university exam may depend upon his/her IQ, mother's qualification, father's qualification, parents income, number of hours of studies, etc.
- Whenever we are interested in studying the joint effect of two or more variables on a single variable, multiple correlation gives the solution of our problem.
- For example, sometimes in data analysis, we have certain factors which are influenced by large number of variables. For instance academic achievement is affected by intelligence, number of hours studied, extra coaching, socio-economic status, etc. We can find out the correlation between academic achievement with various other factors as mentioned above using *Multiple Correlation*.

Multiple Correlation Coefficient

Multiple correlation coefficient is defined as the simple product moment correlation coefficient between observed values and the predicted values of a (response) variable under multivariate set-up.

If x_1 is expressed as a multiple linear regression equation of $x_2, x_3, \dots, x_p$, then the corresponding multiple correlation coefficient will be denoted by $r_{1.23\dots p}$, where

$$r_{1.23\dots p} = \frac{\text{Cov}(x_1, X_{1.23\dots p})}{\sqrt{\text{Var}(x_1)}\sqrt{\text{Var}(X_{1.23\dots p})}} \quad (20)$$

Derivation of the Expression for $r_{1.23\dots p}$

1.

$$\text{Cov}(X_{1.23\dots p}, e_{1.23\dots p}) = 0 \quad (21)$$

Proof.

$$\begin{aligned} \text{Cov}(X_{1.23\dots p}, e_{1.23\dots p}) &= \frac{1}{n} \sum_{\alpha=1}^n (X_{1.23\dots p} - \bar{X}_{1.23\dots p})(e_{1.23\dots p} - \bar{e}_{1.23\dots p}) \\ &= \frac{1}{n} \sum_{\alpha=1}^n (X_{1.23\dots p} - \bar{X}_{1.23\dots p})e_{1.23\dots p} \quad (\text{from (18)}) \\ &= \frac{1}{n} \sum_{\alpha=1}^n [\{\hat{a}_2(x_{2\alpha} - \bar{x}_2) + \hat{a}_3(x_{3\alpha} - \bar{x}_3) + \dots + \hat{a}_p(x_{p\alpha} - \bar{x}_p)\} \\ &\quad \{(x_{1\alpha} - \bar{x}_1) - \hat{a}_2(x_{2\alpha} - \bar{x}_2) - \hat{a}_3(x_{3\alpha} - \bar{x}_3) - \dots - \hat{a}_p(x_{p\alpha} - \bar{x}_p)\}] \\ &= \frac{1}{n} \sum_{\alpha=1}^n \left[\left\{ \sum_{j=2}^p \hat{a}_j(x_{j\alpha} - \bar{x}_j) \right\} \left((x_{1\alpha} - \bar{x}_1) - \sum_{k=2}^p \hat{a}_k(x_{k\alpha} - \bar{x}_k) \right) \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{n} \sum_{j=2}^p \hat{a}_j \sum_{\alpha=1}^n \left\{ (x_{1\alpha} - \bar{x}_1)(x_{j\alpha} - \bar{x}_j) - \sum_{k=2}^p \hat{a}_k (x_{k\alpha} - \bar{x}_k)(x_{j\alpha} - \bar{x}_j) \right\} \\
&= \sum_{j=2}^p \hat{a}_j \left(s_{1j} - \sum_{k=2}^p \hat{a}_k s_{jk} \right) = 0 \quad (\text{from (7)})
\end{aligned}$$

□

2.

$$\text{Var}(x_1) = \text{Var}(X_{1.23\dots p}) + \text{Var}(e_{1.23\dots p}) \quad (22)$$

Proof. Follows directly from (21) as

$$\text{Var}(x_1) = \text{Var}(X_{1.23\dots p}) + \text{Var}(e_{1.23\dots p}) + \text{Cov}(X_{1.23\dots p}, e_{1.23\dots p}) \quad \square$$

3. $\text{Cov}(x_1, X_{1.23\dots p}) = \text{Var}(X_{1.23\dots p})$

Proof. We use the linear model equation $x_1 = X_{1.23\dots p} + e_{1.23\dots p}$ and linearity of covariance in first term to get this result.

$$\begin{aligned}
\text{Cov}(x_1, X_{1.23\dots p}) &= \text{Cov}(X_{1.23\dots p} + e_{1.23\dots p}, X_{1.23\dots p}) \\
&= \text{Cov}(X_{1.23\dots p}, X_{1.23\dots p}) + \text{Cov}(X_{1.23\dots p}, e_{1.23\dots p}) \\
&= \text{Var}(X_{1.23\dots p}) \quad (\text{from (21)})
\end{aligned}$$

□

Therefore, using the above observation we have,

$$\begin{aligned}
r_{1.23\dots p} &= \frac{\text{Cov}(x_1, X_{1.23\dots p})}{\sqrt{\text{Var}(x_1)} \sqrt{\text{Var}(X_{1.23\dots p})}} \\
&= \sqrt{\frac{\text{Var}(X_{1.23\dots p})}{\text{Var}(x_1)}} \geq 0 \quad (23)
\end{aligned}$$

Also, since $r_{1.23\dots p}$ is the simple product moment correlation coefficient between x_1 and $X_{1.23\dots p}$, by definition $r_{1.23\dots p} \leq 1$. Therefore, $0 \leq r_{1.23\dots p} \leq 1$.

We can simplify the above equation for multiple correlation coefficient as follows.

$$\begin{aligned}
\text{Var}(e_{1.23\dots p}) &= \frac{1}{n} \sum_{\alpha=1}^n (e_{1.23\dots p} - \bar{e}_{1.23\dots p})^2 \\
&= \frac{1}{n} \sum_{\alpha=1}^n e_{1.23\dots p}^2 \quad (\text{from (18)})
\end{aligned}$$

$$= \frac{1}{n} SSE = \frac{1}{n} \frac{s_1^2 |R|}{|R_{11}|} \quad (\text{from (19)})$$

$$= \text{Var}(x_1) \frac{|R|}{|R_{11}|} \quad (\text{since } s_1^2 = s_{11} = \sum_{\alpha=1}^n (x_{1\alpha} - \bar{x}_1)^2) \quad (24)$$

From (22), we know that,

$$\begin{aligned}\text{Var}(x_1) &= \text{Var}(X_{1.23\dots p}) + \text{Var}(e_{1.23\dots p}) \\ \Rightarrow \text{Var}(X_{1.23\dots p}) &= \text{Var}(x_1) - \text{Var}(x_1) \frac{|R|}{|R_{11}|} \\ &= \text{Var}(x_1) \left(1 - \frac{|R|}{|R_{11}|}\right)\end{aligned}$$

Therefore, the expression for multiple correlation coefficient is

$$r_{1.23\dots p} = \sqrt{\frac{\text{Var}(X_{1.23\dots p})}{\text{Var}(x_1)}} = \sqrt{1 - \frac{|R|}{|R_{11}|}} \quad (25)$$

4. We know that $0 \leq r_{1.23\dots p} \leq 1$ i.e, $0 \leq \sqrt{1 - \frac{|R|}{|R_{11}|}} \leq 1$. This implies that $0 \leq |R| \leq |R_{11}|$.

If we write $|R_{11}|$ as $|R^{(1)}|$, we get the expression $0 \leq |R| \leq |R^{(1)}|$.

Here $R^{(i_1, i_2, \dots, i_k)}$ for some $\{i_1, i_2, \dots, i_k\} \subset \{1, 2, \dots, p\}$, denotes the submatrix of R obtained by deleting i_1 th, i_2 th, $\dots$, i_k th rows and columns. This notation is used in further derivations as well.

Proceeding this way, we get the following expression due to symmetry.

$$0 \leq |R| \leq |R^{(1)}| \leq |R^{(12)}| \leq \dots \leq |R^{(12\dots p-1)}| = r_{pp} = 1$$

5. More the predictor variables more the multiple correlation coefficient.

$$r_{1.23\dots p} \geq r_{1.23\dots p-1} \quad (26)$$

Proof. If x_1 is regressed on $x_2, x_3, \dots, x_p$, then from (23) we have, $r_{1.23\dots p}^2 = \frac{\text{Var}(X_{1.23\dots p})}{\text{Var}(x_1)}$.

Similarly ⁶, if x_1 is regressed on $x_2, x_3, \dots, x_{p-1}$, $r_{1.23\dots p-1}^2 = \frac{\text{Var}(X_{1.23\dots p-1})}{\text{Var}(x_1)}$

Thus it is good enough to prove that $\text{Var}(X_{1.23\dots p}) \geq \text{Var}(X_{1.23\dots p-1})$.

But from (22) we have, $\text{Var}(X_{1.23\dots p}) + \text{Var}(e_{1.23\dots p}) = \text{Var}(x_1) = \text{Var}(X_{1.23\dots p-1}) + \text{Var}(e_{1.23\dots p-1})$.

Hence it is sufficient to prove that $\text{Var}(e_{1.23\dots p}) \leq \text{Var}(e_{1.23\dots p-1})$

$$\begin{aligned}\text{Var}(e_{1.23\dots p}) &= \frac{1}{n} SSE = \frac{1}{n} \min \sum_{\alpha=1}^n e_{1.23\dots p\alpha}^2 \\ &= \frac{1}{n} \min \sum_{\alpha=1}^n (x_{1\alpha} - x_{1.23\dots p\alpha})^2 \\ &= \frac{1}{n} \min_{a_0, a_2, \dots, a_p} \sum_{\alpha=1}^n \left\{ (x_{1\alpha} - a_0) - \sum_{j=2}^p a_j (x_{j\alpha} - \bar{x}_j) \right\}^2\end{aligned}$$

⁶ Although this relationship is explicitly derived for x_1 regressed on $x_2, x_3, \dots, x_p$, an analogous expression holds for any arbitrary choice of dependent and predictor variables through a suitable selection and permutation of indices. This general applicability is assumed hereafter.

$$\begin{aligned}
&\leq \frac{1}{n} \min_{a_0, a_2, \dots, a_{p-1}} \sum_{\alpha=1}^n \left\{ (x_{1\alpha} - a_0) - \sum_{j=2}^{p-1} a_j (x_{j\alpha} - \bar{x}_j) \right\}^2 \\
&= \text{Var}(e_{1.23\dots p-1}) \\
&\therefore \text{Var}(e_{1.23\dots p}) \leq \text{Var}(e_{1.23\dots p-1})
\end{aligned}$$

□

Partial Correlation Coefficient

When we are interested in the correlation between two variables, eliminating the effects of the rest of the variables (under multivariate set-up), we use *partial correlation coefficient*.

Example

Consider cholesterol level and bank balance for adults. A positive correlation between these two factors is often observed. That is, as the bank balance increases, cholesterol level also increases. But this is not a correct relationship as Cholesterol level can also increase as age increases. Also as age increases, the bank balance may also increase because a person can save from his salary over the years. Thus there is age factor which influences both cholesterol level and bank balance. Suppose we want to know only the correlation between cholesterol and bank balance without the age influence, we could take persons from the same age group and thus control age, but if this is not possible we can statistically control the age factor and thus remove its influence on both cholesterol and bank balance. This if done is called partial correlation. That is, we can use partial correlation for doing the same.

Definition

Let us define the linear models to be $x_1 = X_{1.34\dots p} + e_{1.34\dots p}$ and $x_2 = X_{2.34\dots p} + e_{2.34\dots p}$. Then the partial correlation of x_1 and x_2 eliminating the effect of other variables $x_3, x_4, \dots, x_p$, denoted as $r_{12.34\dots p}$ is defined as follows

$$r_{12.34\dots p} = \frac{\text{Cov}(e_{1.34\dots p}, e_{2.34\dots p})}{\sqrt{\text{Var}(e_{1.34\dots p})} \sqrt{\text{Var}(e_{2.34\dots p})}} \quad (27)$$

where,

$$\begin{aligned}
X_{1.34\dots p} &= \bar{x}_1 + \sum_{j=3}^p \hat{b}_j^{(2)} (x_j - \bar{x}_j), \quad \hat{b}_j^{(2)} = -\frac{s_1}{s_j} \frac{R_{1j}^{(2)}}{R_{11}^{(2)}}, \quad j = 3, 4, \dots, p \\
X_{2.34\dots p} &= \bar{x}_2 + \sum_{j=3}^p \hat{b}_j^{(1)} (x_j - \bar{x}_j), \quad \hat{b}_j^{(1)} = -\frac{s_2}{s_j} \frac{R_{2j}^{(1)}}{R_{22}^{(1)}}, \quad j = 3, 4, \dots, p
\end{aligned}$$

⁷Here we have just taken $\hat{a}_p = 0$. This may or may not be the least square estimator of a_p . Hence by definition this approximation is greater than or equal to SSE.

Derivation of the Expression for $r_{12.34\dots p}$

$$\text{Cov}(e_{1.34\dots p}, x_j) = 0, \quad \forall j = 3, 4, \dots, p^8 \quad (28)$$

Proof.

$$\begin{aligned} \text{Cov}(e_{1.34\dots p}, x_j) &= \frac{1}{n} \sum_{\alpha=1}^n (e_{1.34\dots p\alpha} - \bar{e}_{1.34\dots p})(x_{j\alpha} - \bar{x}_j) \\ &= \frac{1}{n} \sum_{\alpha=1}^n e_{1.34\dots p\alpha} (x_{j\alpha} - \bar{x}_j) \quad (\text{from (18)}) \\ &= \frac{1}{n} \sum_{\alpha=1}^n (x_{1\alpha} - X_{1.34\dots p\alpha})(x_{j\alpha} - \bar{x}_j) \\ &= \frac{1}{n} \sum_{\alpha=1}^n \left[\left(x_{1\alpha} - \bar{x}_1 - \sum_{k=3}^p \hat{b}_k^{(2)} (x_{k\alpha} - \bar{x}_k) \right) (x_{j\alpha} - \bar{x}_j) \right] \\ &= \frac{1}{n} \left\{ \sum_{\alpha=1}^n (x_{1\alpha} - \bar{x}_1)(x_{j\alpha} - \bar{x}_j) - \sum_{k=3}^p \hat{b}_k^{(2)} \sum_{\alpha=1}^n (x_{k\alpha} - \bar{x}_k)(x_{j\alpha} - \bar{x}_j) \right\} \\ &= \frac{1}{n} \left(s_{1j} - \sum_{k=3}^p \hat{b}_k^{(2)} s_{kj} \right) \\ &= 0 \quad (\text{from (7)}) \end{aligned}$$

□

Thus we have,

$$\begin{aligned} \text{Cov}(e_{1.34\dots p}, e_{2.34\dots p}) &= \text{Cov} \left(e_{1.34\dots p}, x_2 - \bar{x}_2 - \sum_{j=3}^p \hat{b}_j^{(1)} (x_j - \bar{x}_j) \right) \\ &= \text{Cov}(e_{1.34\dots p}, x_2) - \sum_{j=3}^p \hat{b}_j^{(1)} \text{Cov}(e_{1.34\dots p}, x_j)^9 \\ &= \text{Cov}(e_{1.34\dots p}, x_2) \quad (\text{from (28)}) \\ &= \text{Cov} \left(x_1 - \bar{x}_1 - \sum_{j=3}^p \hat{b}_j^{(2)} (x_j - \bar{x}_j), x_2 \right) \\ &= \text{Cov}(x_1, x_2) - \sum_{j=3}^p \hat{b}_j^{(2)} \text{Cov}(x_2, x_j) \\ &= \frac{1}{n} s_{12} + \frac{1}{n} \sum_{j=3}^p \left(\frac{s_1}{s_j} \frac{R_{1j}^{(2)}}{R_{11}^{(2)}} \right) s_{2j} \end{aligned}$$

⁸This is one of the most intuitively beautiful results from all of these derivations. Basically it says that the linear relation between a predictor and residuals is zero. Of course, if it were not the case we could have better adjusted the partial regression coefficient of that predictor to reduce the linearity between them.

$$\begin{aligned}
&= \frac{1}{n} \left\{ s_{12} + \sum_{j=3}^p \frac{s_1}{s_j} \frac{R_{1j}^{(2)}}{R_{11}^{(2)}} (s_2 s_j r_{2j}) \right\} \\
\Rightarrow \text{Cov}(e_{1.34\dots p}, e_{2.34\dots p}) &= \frac{1}{n} \left\{ s_1 s_2 r_{12} + \sum_{j=3}^p s_1 s_2 \frac{R_{1j}^{(2)}}{R_{11}^{(2)}} r_{2j} \right\} \\
&= \frac{s_1 s_2}{n R_{11}^{(2)}} \left[r_{12} R_{11}^{(2)} + \sum_{j=3}^p r_{2j} R_{1j}^{(2)} \right] \\
&= -\frac{s_1 s_2}{n R_{11}^{(2)}} \cdot R_{12}
\end{aligned}$$

The last equality follows since

$$R_{12} = (-1)^{1+2} \begin{vmatrix} r_{21} & r_{23} & \dots & r_{2p} \\ r_{31} & 1 & \dots & r_{3p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p3} & \dots & 1 \end{vmatrix} = - \left(r_{21} R_{11}^{(2)} + r_{23} R_{13}^{(2)} + r_{24} R_{14}^{(2)} + \dots + r_{2p} R_{1p}^{(2)} \right)$$

We also have from the derivation of (24) that,

$$\text{Var}(e_{1.34\dots p}) = \frac{s_1^2 |R^{(2)}|}{n R_{11}^{(2)}} \quad \text{and} \quad \text{Var}(e_{2.34\dots p}) = \frac{s_2^2 |R^{(1)}|}{n R_{22}^{(1)}}$$

Thus the final expression for partial correlation coefficient is

$$\begin{aligned}
r_{12.34\dots p} &= \frac{\text{Cov}(e_{1.34\dots p}, e_{2.34\dots p})}{\sqrt{\text{Var}(e_{1.34\dots p})} \sqrt{\text{Var}(e_{2.34\dots p})}} \\
&= \frac{-\frac{s_1 s_2 R_{12}}{n R_{11}^{(2)}}}{\sqrt{\frac{s_1^2 |R^{(2)}|}{n |R_{11}^{(2)}|}} \sqrt{\frac{s_2^2 |R^{(1)}|}{n |R_{22}^{(1)}|}}} \\
&= \frac{-\frac{R_{12}}{R_{11}^{(2)}}}{\sqrt{\frac{|R^{(2)}| |R^{(1)}|}{R_{11}^{(2)} R_{22}^{(1)}}}} \\
&= \frac{-R_{12}}{\sqrt{|R^{(2)}| |R^{(1)}|}} \quad (\text{since } R_{22}^{(1)} = R_{11}^{(2)}) \\
&= \frac{-R_{12}}{\sqrt{R_{11} R_{22}}} \tag{29}
\end{aligned}$$

The last equality follows as $|R^{(1)}| = R_{11}$ and $|R^{(2)}| = R_{22}$.

⁹Here we have used the simple fact that $\text{Cov}(aX + b, Y) = a \text{Cov}(X, Y)$. In particular $\text{Cov}(b, Y) = 0$. Here $\bar{x}_j$'s are constants for every $j = 1, 2, \dots, p$.

Remarks

1. We know, $\hat{a}_j = b_{1j.2,\dots,j-1,j+1,\dots,p}$ for every $j = 2, 3, \dots, p$.

So, $b_{12.34\dots p} = -\frac{s_1}{s_2} \frac{R_{12}}{R_{11}}$ and $b_{21.34\dots p} = -\frac{s_2}{s_1} \frac{R_{21}}{R_{22}} = -\frac{s_2}{s_1} \frac{R_{12}}{R_{22}}$; $R_{12} = R_{21}$ since R is symmetric.

Therefore,

$$b_{12.34\dots p} \times b_{21.34\dots p} = \frac{R_{12}^2}{R_{11}R_{22}} = r_{12.34\dots p}^2$$

2. Let $r_{1(2.34\dots p)}$ denote the correlation between x_1 and the residual of x_2 given $x_3, x_4, \dots, x_p$. Then

$$r_{1(2.34\dots p)}^2 \leq r_{12.34\dots p}^2$$

Proof. We have,

$$\begin{aligned} \text{Cov}(x_1, e_{2.34\dots p}) &= \text{Cov}\left(x_1 - \bar{x}_1 - \sum_{j=3}^p \hat{a}_j(x_j - \bar{x}_j), e_{2.34\dots p}\right) && \text{(from (28))} \\ &= \text{Cov}(x_1 - X_{1.34\dots p}, e_{2.34\dots p}) \\ &= \text{Cov}(e_{1.34\dots p}, e_{2.34\dots p}) \end{aligned}$$

Therefore, we have

$$\begin{aligned} r_{1(2.34\dots p)} &= \text{Cor}(x_1, e_{2.34\dots p}) = \frac{\text{Cov}(x_1, e_{2.34\dots p})}{\sqrt{\text{Var}(x_1)}\sqrt{\text{Var}(e_{2.34\dots p})}} && \text{(from definition)} \\ \Rightarrow r_{1(2.34\dots p)}^2 &= \frac{\text{Cov}^2(x_1, e_{2.34\dots p})}{\text{Var}(x_1) \text{Var}(e_{2.34\dots p})} \\ &= \frac{\text{Cov}^2(e_{1.34\dots p}, e_{2.34\dots p})}{\text{Var}(x_1) \text{Var}(e_{2.34\dots p})} \\ &\leq \frac{\text{Cov}^2(e_{1.34\dots p}, e_{2.34\dots p})}{\{\text{Var}(x_1) - \text{Var}(X_{1.34\dots p})\} \text{Var}(e_{2.34\dots p})} \\ &= \frac{\text{Cov}^2(e_{1.34\dots p}, e_{2.34\dots p})}{\text{Var}(e_{1.34\dots p}) \text{Var}(e_{2.34\dots p})} && \text{(from (22))} \\ &= \text{Cor}^2(e_{1.34\dots p}, e_{2.34\dots p}) \\ \therefore r_{1(2.34\dots p)}^2 &\leq r_{12.34\dots p}^2 \end{aligned}$$

□